

Ritik Bhardwaj

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Projects

Chat Intent Discovery using NLP Embeddings | Scikit-learn, NLTK, Gensim, Transformers May' 26

- Designed an unsupervised NLP pipeline for automatic chat intent discovery by preprocessing customer-support conversations and grouping semantically similar messages without labeled data.
- Integrated multiple text representation techniques including TF-IDF, Word2Vec, GloVe, and Sentence-BERT embeddings to compare semantic understanding and clustering performance across traditional and transformer-based NLP models.
- Applied clustering algorithms such as K-Means, DBSCAN, and HDBSCAN, and evaluated cluster quality using Silhouette Score and Davies-Bouldin Score for effective intent separation.
- Visualized high-dimensional semantic embeddings using PCA and t-SNE to analyze intent separation, semantic similarity, and hidden conversational patterns in large-scale chat datasets.
- **GitHub repository:** Chat Intent Detection

Neural Network Framework from Scratch | Python, NumPy, Matplotlib, OOP Apr' 26

- Built a modular deep learning framework from scratch using Python and NumPy without using high-level libraries like TensorFlow or PyTorch, implementing the core mathematical foundations of neural networks manually.
- Implemented forward propagation, backpropagation, Dense layers, ReLU and Softmax activations, Categorical Cross-Entropy loss, and He initialization using efficient matrix-based computations.
- Developed and compared multiple optimization algorithms including SGD, Momentum, Nesterov Accelerated Gradient (NAG), AdaGrad, RMSProp, and Adam along with learning-rate decay and ReduceLROnPlateau scheduling techniques.
- Improved model generalization and training stability by implementing L1/L2 regularization and inverted Dropout regularization directly inside the custom training pipeline.
- **GitHub repository:** Neural Network from Scratch

Training

Data Structure and Algorithm | Certificate Jun' 25 - Jul' 25

- Completed an intensive and structured training program focused on core and advanced DSA. The program emphasized concept clarity, coding efficiency, and real-world problem-solving techniques.
- Gained experience in advanced topics such as Trees, Graphs, Dynamic Programming, Recursion, and Greedy Algorithms. Learned to apply these concepts to optimize solutions for complex computational problems.
- Strengthened analytical and problem-solving skills through regular coding challenges and timed assessments, improving logical thinking, algorithmic design, and debugging efficiency under competitive conditions.
- Practiced and solved 500+ problems on platforms like LeetCode and GeeksforGeeks. Built consistency in problem-solving while improving accuracy, speed, and code optimization skills.

Certificates

Improving Deep Neural Networks May' 26

Coursera — [Certificate link](#)

Neural Networks and Deep Learning Apr' 26

Coursera — [Certificate link](#)

Machine learning with Python Apr' 26

Coursera — [Certificate link](#)

Skills

Languages: C++, Python, SQL

Libraries/Frameworks: TensorFlow, PyTorch, Keras, NLTK, NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn

Tools/Platforms: MySQL, Git, Linux, AWS, n8n

Soft Skills: Analytical Thinking, Problem-Solving, Team Collaboration, Time Management

Education

Lovely Professional University 2023 – 2027

Bachelor of Technology - Computer Science and Engineering — CGPA: 8.76 Phagwara, Punjab

Army Public School Dhaula Kuan 2021 – 2022

12th — Percentage: 79% Dhaula Kuan, Delhi

Army Public School Bareilly Cantt 2019 – 2020

10th — Percentage: 91.6% Bareilly, Uttar Pradesh